

TQNNs and ZX-Calculus

The preceding sections have established that categorical string diagrams—from DisCoCat’s pregroup derivations (??) through enriched hom-values (??) to topos-theoretic transfer (??)—provide a unified diagrammatic language for case-theoretic reasoning. This section extends the framework into its natural quantum generalization: topological quantum neural networks (TQNNs), the ZX-calculus, and sheaf-theoretic quantum semantic communication. The central observation is that the same monoidal-categorical architecture that underlies DisCoCat string diagrams also underlies quantum circuits, TQFT cobordisms, and quantum information flow—making active inference on case-marked relational structure a quantum topological computation.

QNNs as Spin-Networks

TQFT as the Forward Pass: Reshetikhin–Turaev Invariants Compute Network Amplitudes

Marcianò, Fields, and Glazebrook show that quantum neural networks (QNNs) admit a topological reformulation using